Ashfaq Baig

Senior Data Scientist / Machine Learning Engineer

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Results-driven Data Scientist with 7+ years of hands-on experience in Machine Learning, Generative AI, NLP, and Deep Learning. Proficient in designing and deploying end-to-end AI solutions with expertise in Python, Azure, and edge computing. Adept at solving real-world business challenges using cutting-edge AI/ML techniques and delivering scalable models in production environments.

TECHNICAL SKILLS

Languages & Frameworks: Python, C, C++, Java, JavaScript, HTML/CSS, SQL, R, Azure, Docker,Image Processing

Packages: Scikit-Learn, NumPy, SciPy, Plotly, Pandas, Matplotlib, OpenVino, OpenCV, TensorFlow, Keras,Fast API

Statistics/ML: Linear/Logistics Regression, Random Forest, Gradient Boosted Trees, K-NN, ANN, Generative AI

KEY SKILLS

- Optimization of models
- Data Mining
- Machine Learning
- EDA
- Data Structures
- Data Visualization
- Computer Vision
- Deep Learning

EDUCATION

	Degree / Examination	Board/University	From Year	To Year	% of marks	Part time/Full time/Distance Learning
Graduation	Bachelor	JNTUK	2012	2016	7.76 CGPA	Full time
XII th	Intermediate	State	2010	2012	89.5%	Full time
X th	10th	State	2009	2010	80.5%	Full time

CERTIFICATIONS

- Master Certification in Artificial Intelligence | Simplilearn
- Deep Learning | Simplilearn
- Data Science | Simplilearn
- Machine Learning | Simplilearn

PROFESSIONAL EXPERIENCE

Senior Data Scientist

Fractal Analytics **March 2024 – Present Bengaluru, IN**

Project: Climate Risk Assessment GenAI Assistant (CCRA) (Client: ANZ Bank)

- Architected and productionized a GenAI system to automate climate risk document analysis for financial institutions, reducing manual review time by ~85% with auditability and compliance controls.

- Built a custom orchestration framework on Vertex AI for Gemini 2.5 Pro enabling parallel multi-document analysis, intelligent page-splitting, and summary aggregation for higher throughput and lower latency.
- Implemented a retrieval pipeline using embedding similarity with cosine search to surface the top-k relevant content for grounded, context-aware answers.
- Delivered a Q&A application with two-stage answer validation, compliance checks, and audit ready rationales, including caching, session management, and detailed logging.

Technical stack — Climate Risk Assessment GenAI Assistant (CCRA)

- Modeling: DexiNed architecture in PyTorch; data augmentation; edge/contour evaluation.
- Languages/SDKs: Python, Vertex AI SDK, REST/JSON, OAuth/service accounts.
- LLM/GenAI: Gemini 2.5 Pro on Vertex AI, prompt templates, structured outputs.
- Retrieval: Embedding generation, vector similarity with cosine distance, reranking strategy.
- Document processing: PyMuPDF (Fitz) for PDF parsing/OCR fallback; text normalization and chunking.
- App/Backend: Streamlit for UI, FastAPI microservices for orchestration and APIs, Dockerized services.
- Cloud: GCP Cloud Run, Cloud Storage.

Project: Optical Blade Layout (Client: Siemens Gamesa)

- Trained a DexiNed edge detection model reaching an F1-score of 82% and optimized with FP16 post-training quantization and pruning for efficient inference.
- Deployed on NVIDIA Triton Inference Server achieving ~4× speedup in production through dynamic batching and optimized backends for low-latency serving.
- Contributed to CI/CD pipeline and Azure IoT Hub–to–edge deployments with monitoring and safe rollbacks.

Technical stack — Optical Blade Layout

- Modeling: DexiNed architecture in PyTorch; data augmentation; edge/contour evaluation.
- Optimization: FP16 quantization, structured pruning, model export for serving backends.
- Serving: Triton Inference Server with dynamic batching, gRPC/HTTP endpoints, model repository.[5][2]
- Servicesasync workers
- MLOps: Docker images, CI/CD, infra-as-code patterns, metrics.
- Edge/Cloud: Azure IoT Hub to edge nodes; deployment orchestration and telemetry integration.

Data Scientist

Quest Global Engineering Services Private Limited **Jan 2021 – March 2024 Bengaluru, IN**

Project: Decision Support-Mooring Operations

- Developed and trained a DNN model for ship stability prediction, combining optimization (SLSQP) and ANN to minimize error to 1.76e-06.
- Applied Bayesian hyperparameter tuning for Neural Networks.
- Generated synthetic data using physics-based systems to train the ANN model and deployed it into the Decision Engine using NumPy and Pandas.
- Utilized SQL for data extraction, summarization, and visualization.
- Managed all phases of the Software Development Life Cycle (SDLC) independently.

Machine Leaning Developer-II

Tardid Technologies Private Limited **Mar 2018 – Dec 2020 Bengaluru, IN**

Project: Real-Time Crack Detection for High-Pressure Vessels

- Analyzed thermal video data and developed solutions for crack detection.
- Built a pipeline for pre-processing, model inference, and evaluation.
- Implemented image processing techniques (masking, de-noising, labeling) using OpenCV.

- Developed a segmentation algorithm (U-Net) for crack detection, achieving 98.2% accuracy on test data.
- Trained the model on GPU and deployed it on Raspberry Pi 3 for real-time inference.
- Conducted a feasibility study using OpenVINO to select hardware (Intel Atom) for model inference.